

Fifth day of class - 9/9

Last time: We talked about the operation $P * Q$, and using that to create an operation $P + Q$ that satisfies the four properties we want to have.

Plan: Today, we will talk about projective space. We'll be working in projective space a lot, and so it's helpful to feel comfortable with it.

Definition: If F is a field, then $\mathbb{P}^n(F)$ is the set of points

$$p = (a_0 : a_1 : \cdots : a_n).$$

We require that not all of the a_i are equal to zero. Also, if $c \neq 0$, then

$$(a_0 : a_1 : \cdots : a_n) = (ca_0 : ca_1 : \cdots : ca_n).$$

(Technically, we could define an equivalence relation on the set of $n + 1$ tuples of elements of F that are not all zero. Two such tuples are equivalence if the corresponding vectors are scalar multiples of each other.)

Example: $n = 1$ and $F = \mathbb{Q}$. Any point in $\mathbb{P}^1(\mathbb{Q})$ has the form

$$(a_0 : a_1)$$

where a_0 and a_1 are not both zero.

If $a_1 \neq 0$, then we can scale the point by $1/a_1$ and get

$$(a_0/a_1 : 1).$$

If $a_1 = 0$, then we have the point $(a_0 : 0)$. Then $a_0 \neq 0$ and this point can be scaled to be $(1 : 0)$.

So every point in $\mathbb{P}^1(\mathbb{Q})$ has the form $(a_0 : 1)$, or is equal to $(1 : 0)$. So $\mathbb{P}^1$ is “regular 1-dimensional space” with one point added.

In $\mathbb{P}^2$, every point has the form $(a_0 : a_1 : a_2)$. If $a_2 \neq 0$, then we can scale to get $(a_0/a_2 : a_1/a_2 : 1)$. If $a_2 = 0$, then we get $(a_0 : a_1 : 0)$. So $\mathbb{P}^2$ is like “regular 2-dimensional space”, with a $\mathbb{P}^1$ added.

In general, each point in $\mathbb{P}^n(F)$ has a unique representation where the last nonzero entry is a 1.

Q: How can we define curves or surfaces inside $\mathbb{P}^n(F)$?

A: In order for a polynomial $f(x_0, \dots, x_n) = 0$ to define a set inside $\mathbb{P}^n(F)$, we need it to be the case that if $(a_0 : a_1 : \cdots : a_n) = (b_0 : b_1 : \cdots : b_n)$ then

$$f(a_0, a_1, \dots, a_n) = 0 \text{ if and only if } f(b_0, b_1, \dots, b_n) = 0.$$

To make this happen, we will assume that f is *homogeneous* - each term in f has the same degree. Then, if $b_i = ca_i$, then

$$f(b_0, b_1, \dots, b_n) = c^{\deg f} f(a_0, a_1, \dots, a_n)$$

and so $f(b_0, \dots, b_n) = 0 \iff f(a_0, \dots, a_n) = 0$.

Example 1: Where do the lines $2x + y + z = 0$ and $3x + 2y + z = 0$ intersect?

We can represent a typical point on the first line by $(x : y : z) = (x : y : -2x - y)$. We can represent a typical point on the second line as $(x : y : z) = (x : y : -3x - 2y)$. So for them to intersect, we need $-2x - y = -3x - 2y$. This gives $x = -y$. So we have

$$(x : y : -2x - y) = ((-y) : y : (-2)(-y) - y) = (-y : y : y) = (-1 : 1 : 1).$$

Example 2: Where do the lines $x + y + 2z = 0$ and $x + y + 3z = 0$ intersect?

Subtracting the two equations we get $z = 0$. Also, $x + y = 0$. Thus, we have $(-y : y : 0) = (-1 : 1 : 0)$.

Example 3: If we take an elliptic curve,

$$E : y^2 = x^3 + Ax + B$$

we can make this equation homogeneous by adding another variable z and using that to make all the terms have the same degree. This gives us

$$E : y^2 z = x^3 + Axz^2 + Bz^3.$$

We can then talk about the points on E in $\mathbb{P}^2(F)$. If $(x : y : z)$ is such a point, and $z \neq 0$, then we can scale and get $(x/z : y/z : 1)$. This gives us precisely the points we had originally. If we set $X = x/z$ and $Y = y/z$, then $Y^2 = X^3 + AX + B$.

On the other hand, if $z = 0$, then we plug this into our curve and get $0 = x^3 + 0 + 0$ and so $x = 0$. Since y must then be nonzero, we can scale it and get the point $(0 : 1 : 0)$. This point is the “point at infinity”, and is the identity of the group law on E . The proper definition of $E(\mathbb{Q})$ is the set of points $P = (x : y : z) \in \mathbb{P}^2(\mathbb{Q})$ that satisfy $y^2 z = x^3 + Axz^2 + Bz^3$.

Why should we work in projective space?

Answer 1: There are a number of annoying situations that arise in the “ordinary plane” that we can remedy by working in projective space. In the ordinary plane, every line can be written in the form $y = mx + b$ *except* those lines that are vertical.

Also, in the ordinary plane, any two lines intersect in exactly one point *unless* they are parallel.

In the projective plane, any two distinct lines intersect in *exactly* one point. If the lines are “parallel”, they intersect at a point at infinity.

(Draw a representation of $\mathbb{P}^2$ on the board.)

Theorem (Bezout's theorem). *If C_1 and C_2 are two distinct curves in $\mathbb{P}^2(\mathbb{C})$ with degree m and degree n , then the curves C_1 and C_2 intersect in exactly mn points (counting multiplicity).*

The projective plane is basically the smallest set that contains the regular plane in which any two lines intersect in exactly one point.

This guarantees that any line intersects an elliptic curve in *exactly* three points (counting multiplicity).

(My algebraic geometry professor in grad school said that projective space is “compact”. If we don't work in projective space then lines or curves or surfaces might have “missing points” that are limits of other points. When we parametrized genus 0 curves last week, we got some missing points. Those missing points go away if we represent our parametrizations as a map from $\mathbb{P}^1$ to points on the conic in projective space.)

Answer 2: We want to be able to reduce rational points modulo a prime p . Let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} = \{0, 1, 2, \dots, p-1\}$. There is no natural map $\mathbb{Q} \rightarrow \mathbb{F}_p$. (In abstract algebra language, there's no nonzero ring homomorphism from $\mathbb{Q}$ to $\mathbb{F}_p$.)

However, there *is* a natural map from $\mathbb{P}^2(\mathbb{Q})$ to $\mathbb{P}^2(\mathbb{F}_p)$. If $(x : y : z)$ is a point in $\mathbb{P}^2(\mathbb{Q})$, we there is a unique (up to sign) rational number d so that dx , dy and dz are all integers that don't have a common factor > 1 .

We have that $(x : y : z) = (dx : dy : dz)$. Then, at least one of $dx \bmod p$, $dy \bmod p$ and $dz \bmod p$ is nonzero mod p and so

$$(dx \bmod p : dy \bmod p : dz \bmod p) \in \mathbb{P}^2(\mathbb{F}_p).$$

For example, if $E : y^2z = x^3 - 36xz^2$, and we have the point $P = (25/4 : -35/8 : 1)$, we can multiply through by 8 and get $P = (50 : -35 : 8)$. Now, if we want to reduce this point modulo 11, we get

$$(50 : -35 : 8) \equiv (6 : 9 : 8) \equiv (9 : 8 : 1) \pmod{11}.$$

If we reduce this point mod 2, we get $(50 : -35 : 8) \equiv (0 : 1 : 0) \pmod{2}$. This is the point at infinity on E .

In general if $P = (a/b : c/d : 1)$ is a point on an elliptic curve $E : y^2 = x^3 + Ax + B$ where $A, B \in \mathbb{Z}$, and p divides b or p divides d , then p divides both b and d and the point P reduced mod p to $(0 : 1 : 0)$.

The points at infinity on a curve in $\mathbb{P}^2$

If C is a curve in $\mathbb{P}^2$ defined by a homogeneous polynomial $f(x, y, z)$, then there are two sorts of points on C .

The points on C that are “affine” points: these are the points $(x : y : 1)$ where $f(x, y, 1) = 0$.

The points on C that are “at infinity”. These are the points $(x : y : 0)$ where $f(x, y, 0) = 0$. These correspond to the “directions” in which the curve heads off to infinity.

The book gives the example of the curve $f(x, y, z) = x^2 - y^2 - z^2$, whose affine points are the hyperbola $x^2 - y^2 = 1$. This hyperbola has two asymptotes, corresponding to $y = \pm x$, and this gives rise to the two points $(1 : 1 : 0)$ and $(1 : -1 : 0)$ at infinity.

What about $f(x, y, z) = x^2 + y^2 - z^2 = 0$? The affine points of this curve are a circle, namely $(x : y : 1)$ where $x^2 + y^2 = 1$. What about the points at infinity? These arise when $x^2 + y^2 = 0$. There are no real points at infinity.

However, Bezout’s theorem tells us that the line $z = 0$ (the line of points at infinity in $\mathbb{P}^2$) must intersect this degree 2 curve in two points. Where are those points?

We can write $y^2 = -x^2$ and taking square roots gives $y = ix$ or $y = -ix$. This gives the points $(x : ix : 0) = (1 : i : 0)$ and $(x : -ix : 0) = (1 : -i : 0)$.

Constructing tangent lines

Q: Suppose that you have a curve C in $\mathbb{P}^2$ defined by the homogeneous polynomial $f(x, y, z)$. What is the equation of the tangent line to C at $(a : b : c)$?

A: We let $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$ be the partial derivatives of f with respect to each of the variables. The expression $\frac{\partial f}{\partial x}$ is obtained by thinking of y and z as constants, and computing the derivative with respect to x .

The equation of the tangent line is

$$\frac{\partial f}{\partial x}(a, b, c)(x - a) + \frac{\partial f}{\partial y}(a, b, c)(y - b) + \frac{\partial f}{\partial z}(a, b, c)(z - c) = 0.$$

But this equation is supposed to be a line. This means it needs to be homogeneous. The constant term is

$$-a \frac{\partial f}{\partial x}(a, b, c) - b \frac{\partial f}{\partial y}(a, b, c) - c \frac{\partial f}{\partial z}(a, b, c).$$

Assuming that (a, b, c) is on the curve, this is always zero. (It follows from a multivariable version of the chain rule.)

Example: Let $f(x, y, z) = y^2 z - x^3 - z^3$ and $P = (2 : 3 : 1)$.

We have

$$\begin{aligned} \frac{\partial f}{\partial x} &= -3x^2 \\ \frac{\partial f}{\partial y} &= 2yz \\ \frac{\partial f}{\partial z} &= y^2 - 3z^2. \end{aligned}$$

Plugging in $x = 2$, $y = 3$ and $z = 1$ we get

$$\frac{\partial f}{\partial x}(0, 1, 1) = -12$$

$$\frac{\partial f}{\partial y}(0, 1, 1) = 6$$

$$\frac{\partial f}{\partial z}(0, 1, 1) = 9 - 3 = 6.$$

So the equation of our line is

$$\begin{aligned} -12(x - 2) + 6(y - 3) + 6(z - 1) &= 0 \\ -12x + 24 + 6y - 18 + 6z - 6 &= 0. \end{aligned}$$

The constant term cancels (at it must)!

We can rewrite this as $-2x + y + z = 0$. The intersection of this line with the curve is where $y = 2x - z$. This gives

$$\begin{aligned} (2x - z)^2 z - x^3 - z^3 &= 0 \\ 4x^2 z - 4xz^2 + z^3 - x^3 - z^3 &= 0 \\ -x^3 + 4x^2 z - 4xz^2 &= 0 \\ -x(x^2 - 4xz + 4z^2) &= 0 \\ -x(x - 2z)^2 &= 0. \end{aligned}$$

We see that $x - 2z$ is squared and so this line intersects the curve at the point $x = 2z$, $y = 2x - z = 2(2z) - z = 3z$ at $(2z : 3z : z) = (2 : 3 : 1)$ with multiplicity two.

The other intersection point is $x = 0$. This gives $y = -z$ and so $(x : y : z) = (0 : -z : z) = (0 : -1 : 1)$.

This is saying that if $P = (2 : 3 : 1)$, then $P * P = (0 : -1 : 1)$ and so $P + P = (0 : 1 : 1)$.

One more example (if time): Can you use modular arithmetic to prove that there are no rational solutions to

$$x^5 + y^5 = 3?$$